Noise-induced shallow circuits and absence of barren plateaus

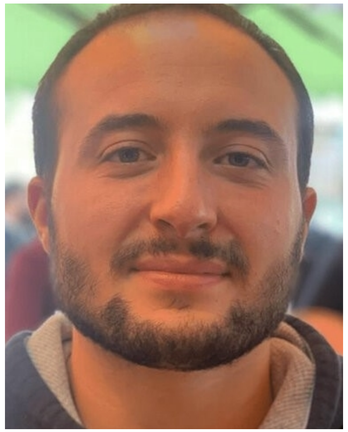
Antonio Anna Mele

arXiv:2403.13927

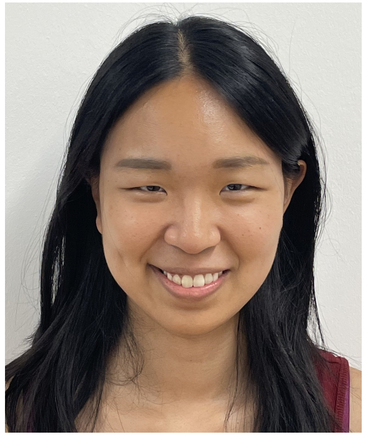


Joint work with:

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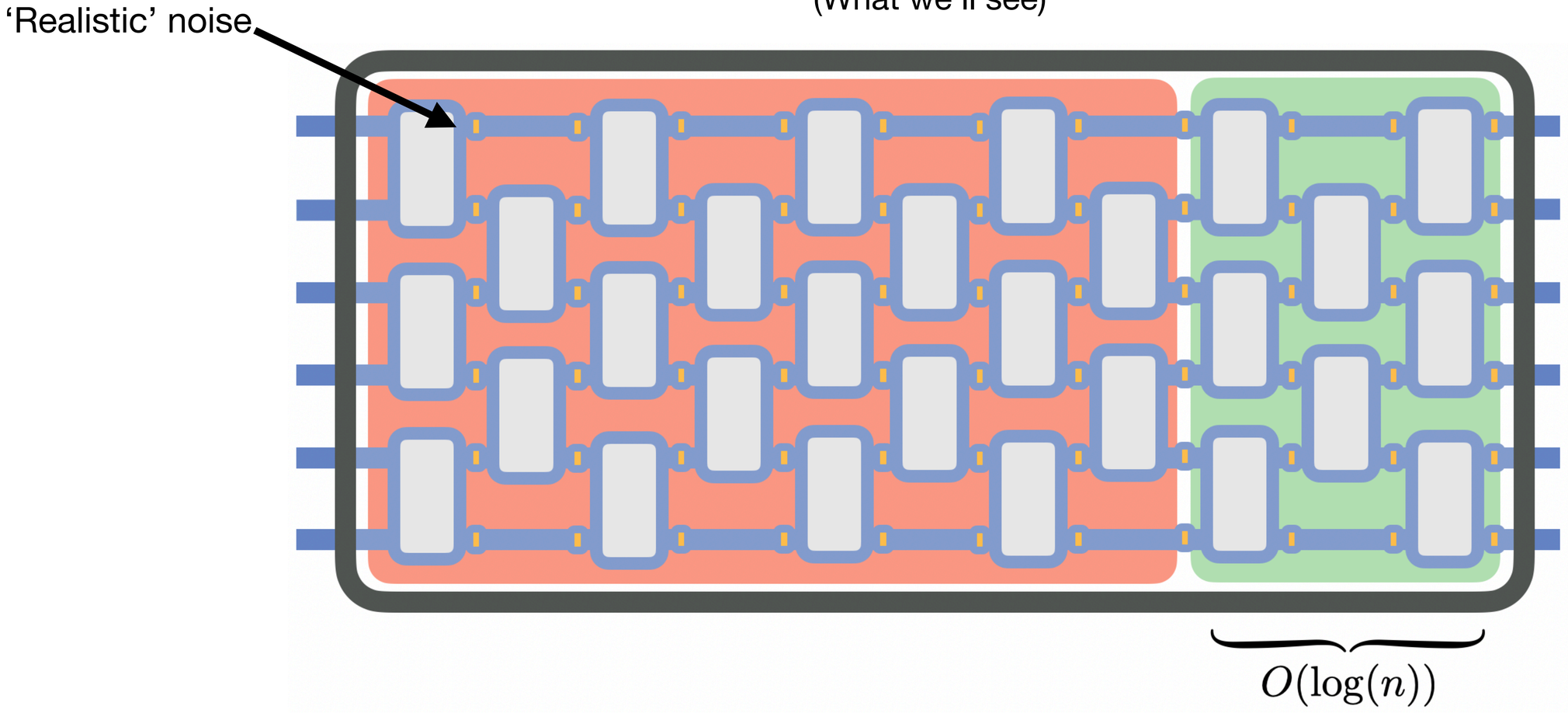
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Spoiler

(What we'll see)



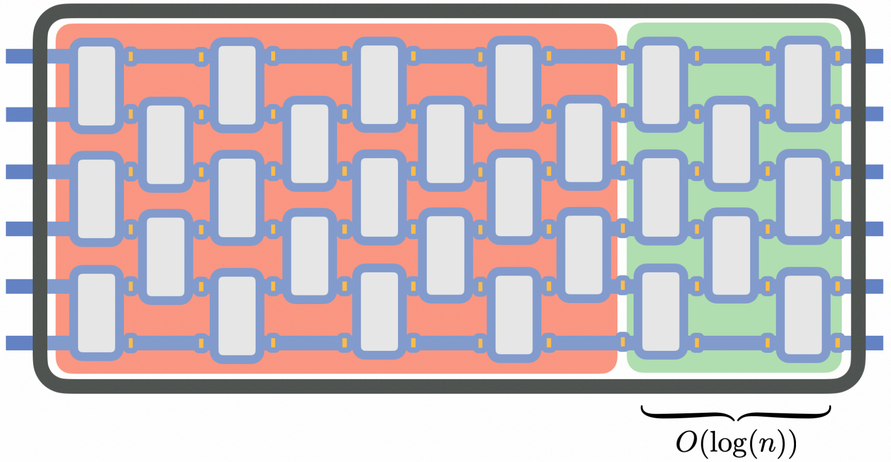
Only the last layers matter to estimate expectation values, for most circuits.

This implies:

- Efficient classical simulation
- Lack of Barren plateaus

Outline

- Noise in quantum circuits
- Effective shallow circuits
- Classical simulation of Pauli expectation values of noisy random circuits
- Barren plateaus



- Understanding **noise impact** in current quantum devices is important for:



- Many previous works modeled noise as solely **depolarizing**.

$$\mathcal{N}(\sigma) = (1-p)\sigma + p\frac{I}{2} \quad \text{with } p \in [0,1]$$

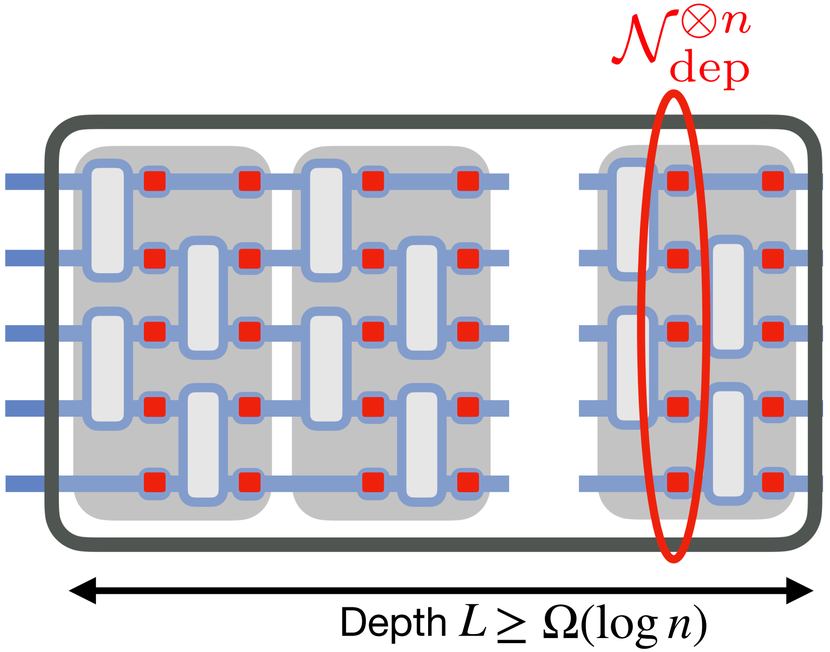
- **Depolarizing noise**, e.g., induces:
  - Barren plateaus in variational quantum algorithms [1],
  - Efficient classical simulation of **'supremacy' sampling experiments** [2].

- But, **tiny departure** from this model could lead to **different conclusions** [3,4].

[1] Noise-induced barren plateaus in variational quantum algorithms. Wang *et al,* Nature Comm. *(2021).*
[2] A polynomial-time classical algorithm for noisy random circuit sampling, Aharonov *et al*., STOC (2023)
[3] Quantum refrigerator, Ben-Or *et al*, QIP (2013).
[4] Effect of non-unital noise on random circuit sampling, Fefferman *et al*, QIP (2023).

- The **depolarizing** assumption, if true, **enormously constrains noisy computation**:

$\Phi_L =$



$$\left\| \Phi_L(\rho_0) - \frac{\mathbb{I}}{2^n} \right\|_1 \leq \exp(-\Theta(L)), \text{ [5]}$$

Quantum computation possible only for $\log(n)$ depth! [6]

$$|\operatorname{Tr}(P\Phi_L(\rho_0))| \leq \exp(-\Theta(L)),$$

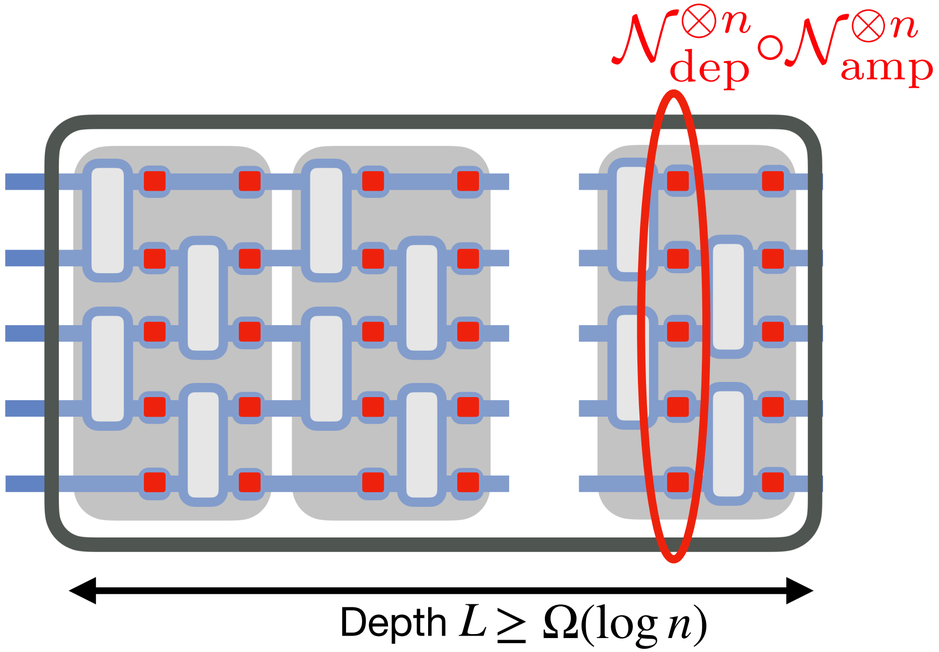
for any Pauli P and input state ρ_0

- Thus, **efficient** classical simulation of **all** deep depolarizing-noisy circuits is “trivial”.

[5] Relative entropy convergence for depolarizing channels, Muller-Hermes *et al,* J. Math. Phys. (2016).
[6] Limitations of noisy reversible computation, Aharonov et al, ArXiv (1996).

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Quantum computation possible only for $\log(n)$ depth! [6]

However, for noises different from depolarizing, not all circuits become trivial after log-depth!

- Real **hardware noise** is **not solely depolarizing** (which is 'unital', i.e., $\mathcal{N}(I) = I$)
- Other noise components present (T1-error, amplitude damping, …) are **non-unital**.

[3] Quantum refrigerator, Ben-Or *et al*, QIP (2013). → **'Non-unital' noisy circuits can be made fault-tolerant, without fresh auxiliary qubits!**

[5] Relative entropy convergence for depolarizing channels, Muller-Hermes *et al,* J. Math. Phys. (2016).
[6] Limitations of noisy reversible computation, Aharonov et al, ArXiv (1996).

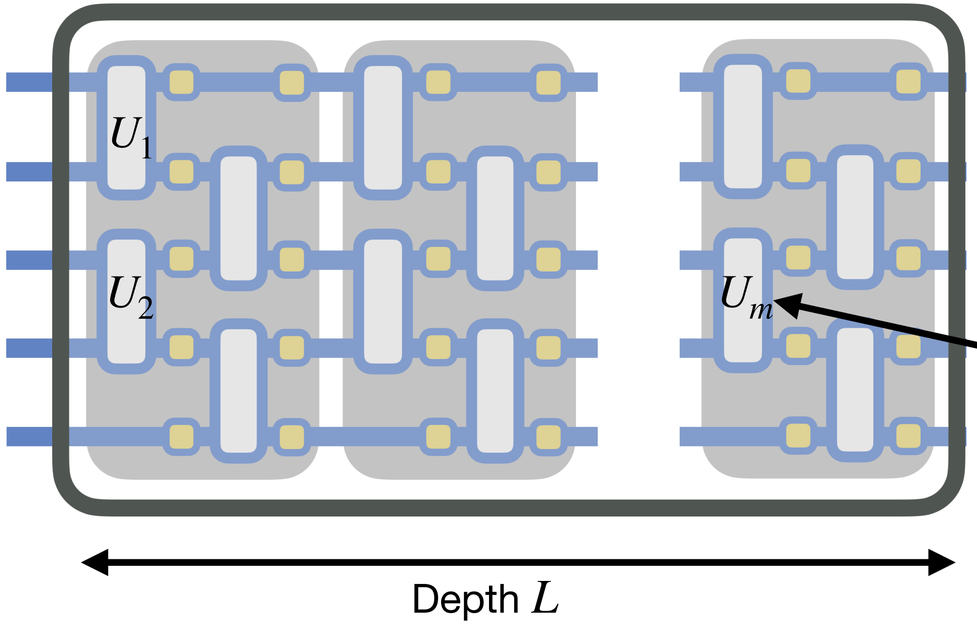
Thus, if the noise is non-unital:
one can carefully engineer the circuit to perform reliable quantum computation [3]

(If BQP≠BPP, Pauli expectation values of **all** non-unital noisy circuits cannot be efficiently estimated classically.)

But, what happens for generic noisy circuits?

(Our work)

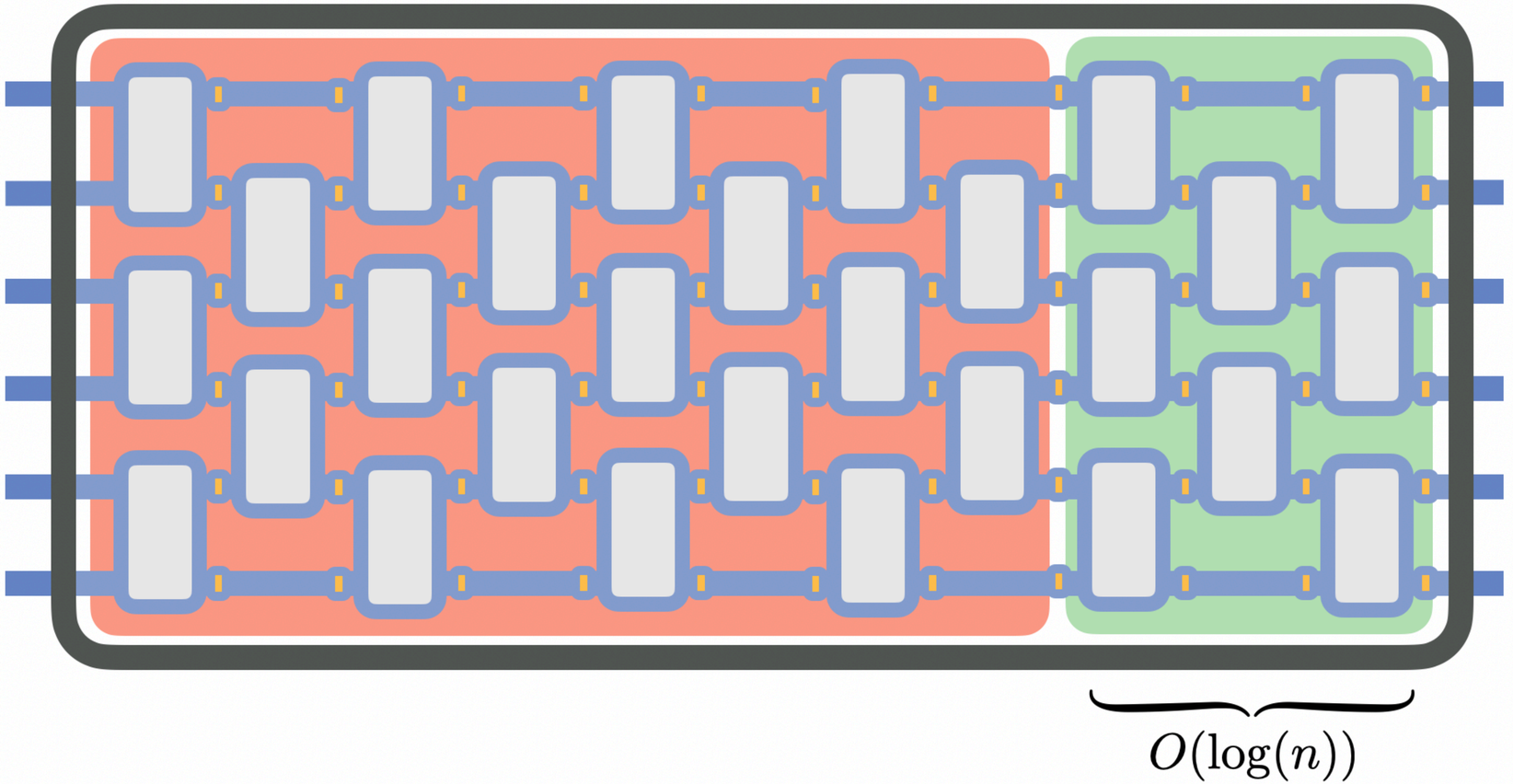
$\Phi_L =$



$\sim$ Haar(U(4))

[3] Quantum refrigerator, Ben-Or *et al*, QIP (2013).

What happens for generic noisy circuits?

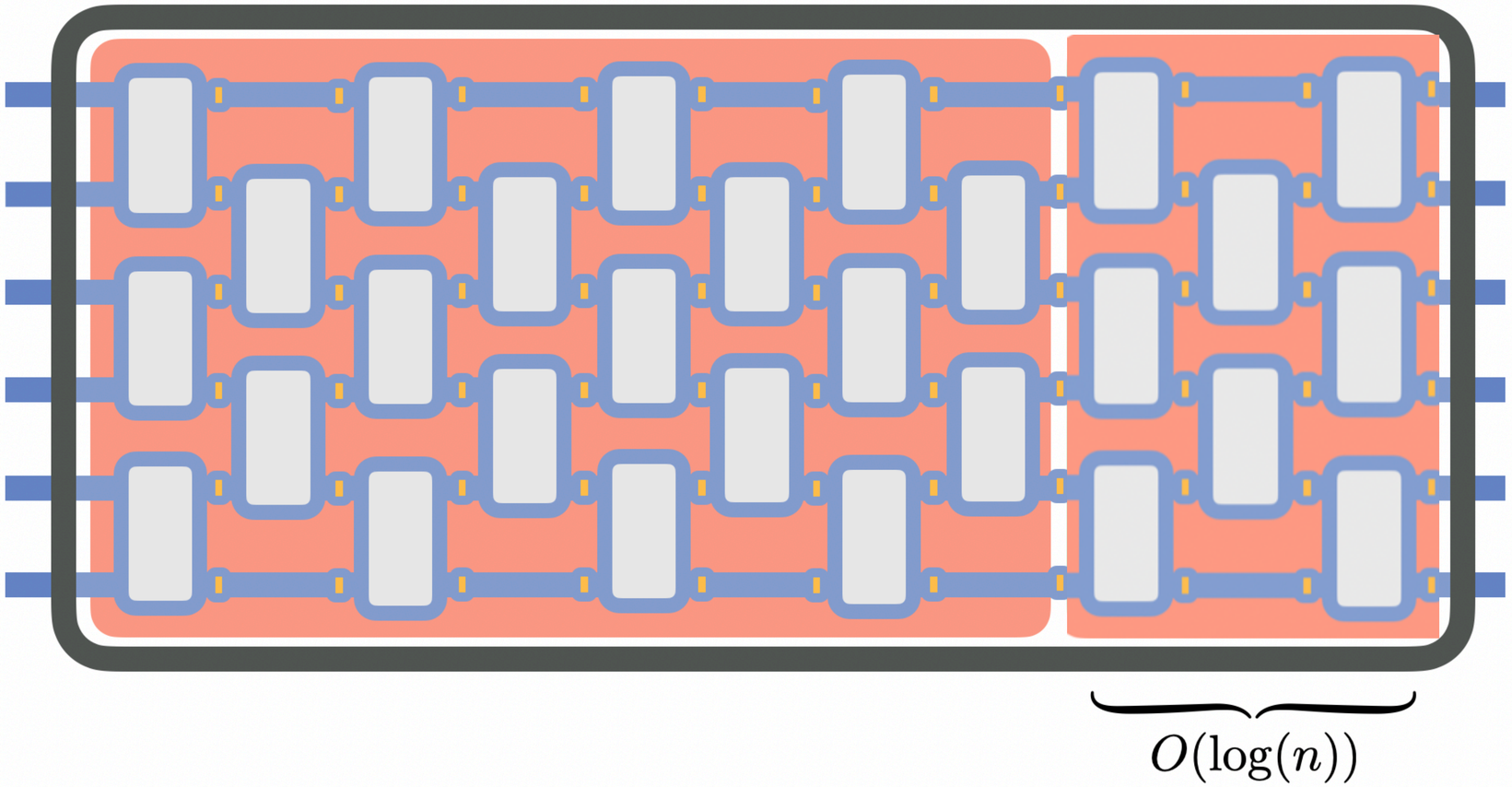


Only the last $\log(n)$ layers matter to estimate expectation values, for most circuits.

Intuition: the content of the first layers is forgotten, if nothing *clever* is done to remember!

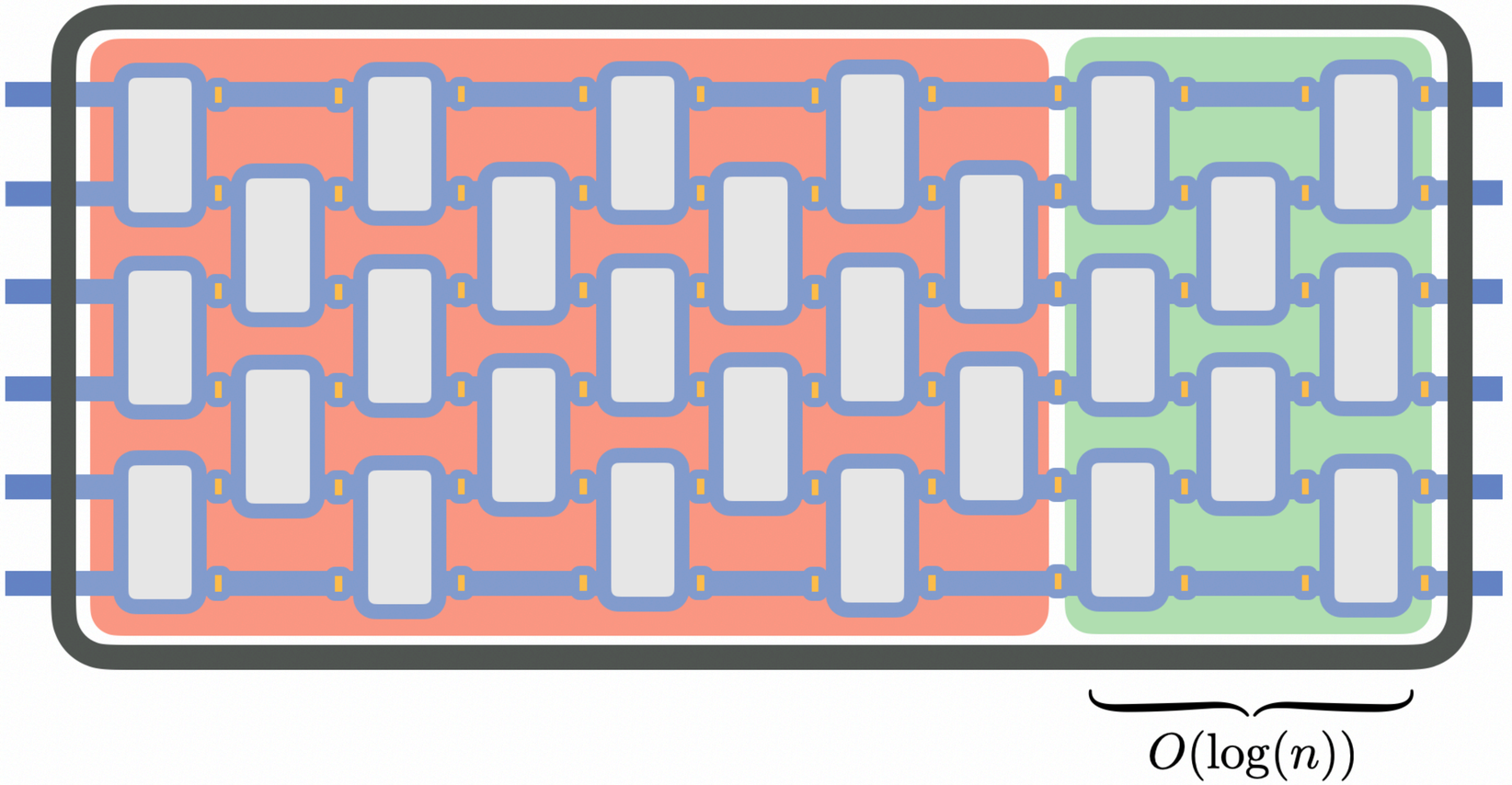
If the noise is **depolarizing**, no layers matter.

(Even for ALL circuits)



If the noise is **non-unital**, the last $O(\log(n))$ do matter!

(for most circuits)



Effective shallow circuits

Proposition.

For any initial state ρ_0 (possibly complex) and observable O,

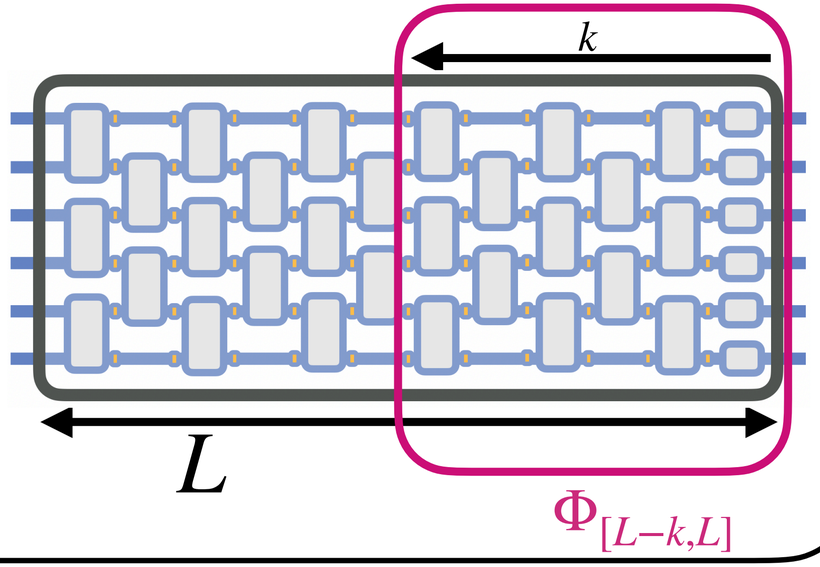
$$\mathbb{E}_\Phi \left[\left| \mathrm{Tr}\left(O\Phi(\rho_0)\right) - \mathrm{Tr}\left(O\Phi_{[L-k,L]}(\sigma_0)\right) \right| \right] \leq \|O\|_\infty \exp(-\Omega(k)).$$

Target expectation value

Last k layers

Any preferred initial state

$\Phi =$



This follows from the following more general theorem:

Theorem.

For any k depth of the circuit Φ_k, states ρ, σ, and Pauli P, we have:

$$\mathbb{E}_{\Phi_k} \left[\left| \mathrm{Tr}(P\Phi_k(\rho)) - \mathrm{Tr}(P\Phi_k(\sigma)) \right| \right] \leq \exp(-\Omega(k + |P|)).$$

Trace distance convergence

Theorem. (Trace distance bound)
For any $L \geq \Omega(n)$ depth of the noisy n-qubits circuit Φ, states ρ, σ, we have:

$$\mathbb{E}_{\Phi} \left\| \Phi(\rho) - \Phi(\sigma) \right\|_1 \leq \exp(-\Omega(L)).$$

- A general result like this, but without $\mathbb{E}_{\Phi}$, cannot be obtained (e.g.) because of Ref. [3]
  (which shows how to leverage non-unital noise to perform fault-tolerant quantum computation without fresh-auxiliary qubits.)

- However, in certain 'high noise' regime, we can obtain it:

Theorem. (High-noise worst-case, informal)
For any $L \geq \Omega(\log(n))$ depth of the (very) noisy n-qubits circuit Φ, states ρ, σ, we have:

$$\left\| \Phi(\rho) - \Phi(\sigma) \right\|_1 \leq \exp(-\Omega(L)).$$

[3] Quantum refrigerator, Ben-Or *et al*, QIP (2013).

Classical simulation of noisy random circuits

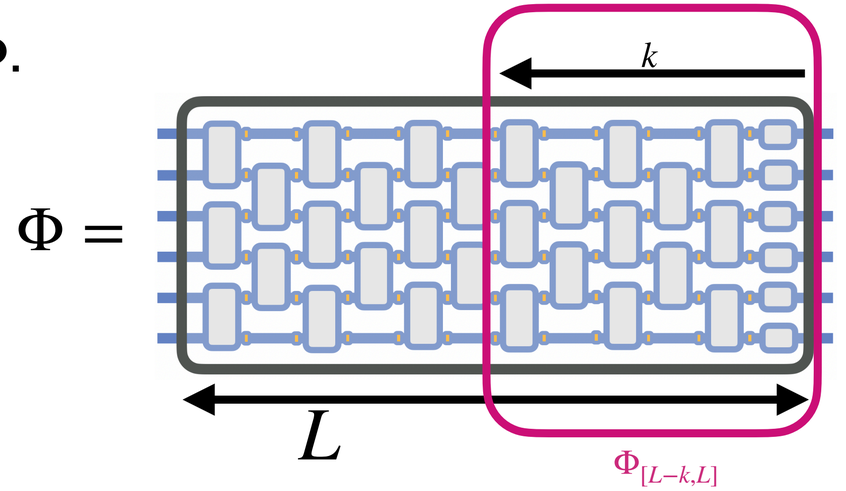
Task: Estimate $\mathrm{Tr}\left(P\Phi(\rho_0)\right)$, with high probability over the choice of Φ.

Solution: It suffices to output $\mathrm{Tr}\left(P\Phi_{[L-k,L]}(|0^n\rangle\langle 0^n|)\right)$.

Previous proposition.

$$\mathbb{E}_\Phi\left[\left|\mathrm{Tr}\left(P\Phi(\rho_0)\right)-\mathrm{Tr}\left(P\Phi_{[L-k,L]}(|0^n\rangle\langle 0^n|)\right)\right|\right]\leq \exp(-\Omega(k+|P|)).$$

Choosing $k=O(\log(\varepsilon^{-1}))$ suffices to have: $\leq \varepsilon$

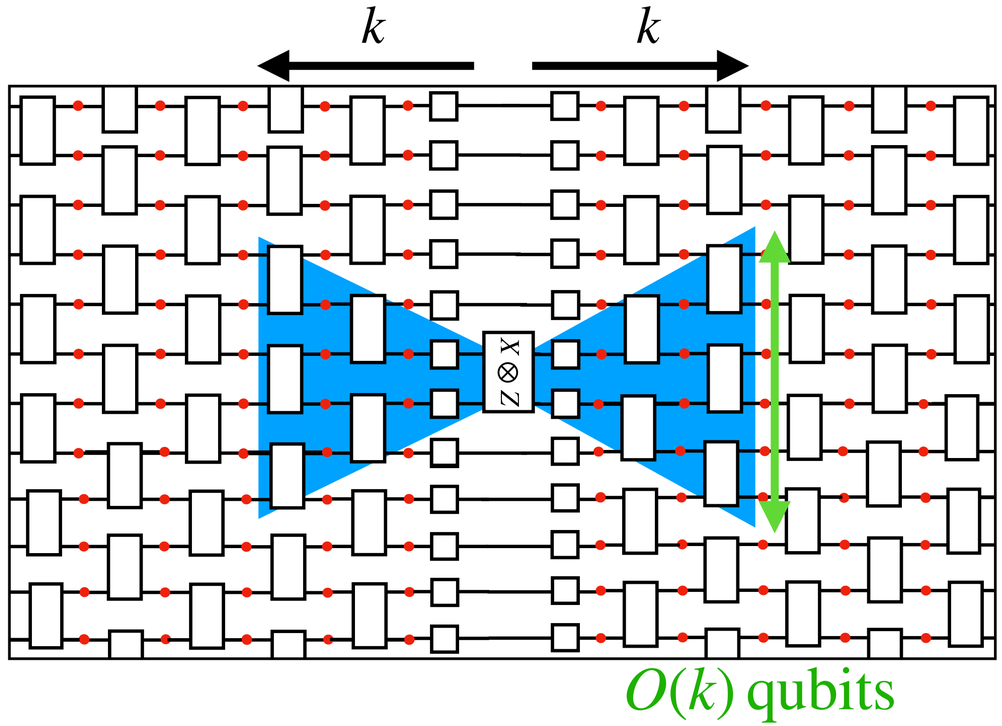


- We have $\mathrm{Tr}\left(P\Phi_{[L-k,L]}(|0^n\rangle\langle 0^n|)\right)=\mathrm{Tr}\left(\Phi_{[L-k,L]}{}^*(P)|0^n\rangle\langle 0^n|\right)$

If P is local, then this is also local

- Thus, this can be computed efficiently for light-cone arguments.

Computational time (for 1D): $\exp(O(k))=\mathrm{poly}(\varepsilon^{-1})$



Classical simulation of noisy random quantum circuits expectation values

Theorem.

Given a randomly sampled noisy quantum circuit, for any depth and any noise, there is a classical algorithm to **estimate local Pauli expectation values** up to ε additive precision, with high probability over the choice of the circuit.

The runtime is $\exp\left(O(\log^{\mathrm{D}}(\varepsilon^{-1}))\right)$, where D is the spatial dimension of the system.



(No dependence by the number of qubits and depth.)

- Thus, for constant ε precision, the algorithm is efficient for any dimensionality.

- For $\varepsilon = \mathrm{poly}(n^{-1})$, the algorithm is efficient for 1D architectures, and quasi-poly for $\mathrm{D} > 1$.

- For **global Pauli** expectation values, output zero succeeds with high probability.

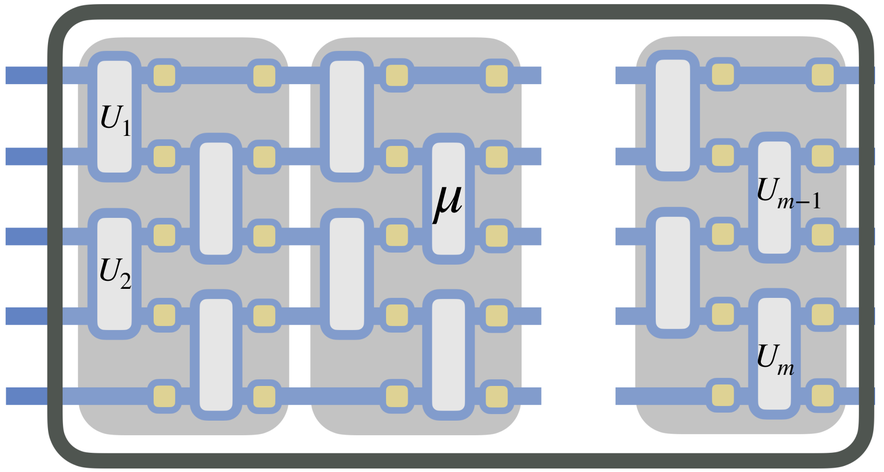
- We also give a **condition to verify the success of the classical simulation**:
  "Check if the Heisenberg evolved observable is $\sim$ proportional to the Identity".

Barren plateaus

- **Variational quantum algorithms** encode the problem into $\min_{U_1,\ldots,U_m} \mathrm{Cost}(\Phi_{U_1,\ldots,U_m})$

- **Barren plateaus (informal)** iff for most choices of $U_1, \ldots, U_m$ the cost landscape is 'flat' (gradient norm very small).

$\Phi =$



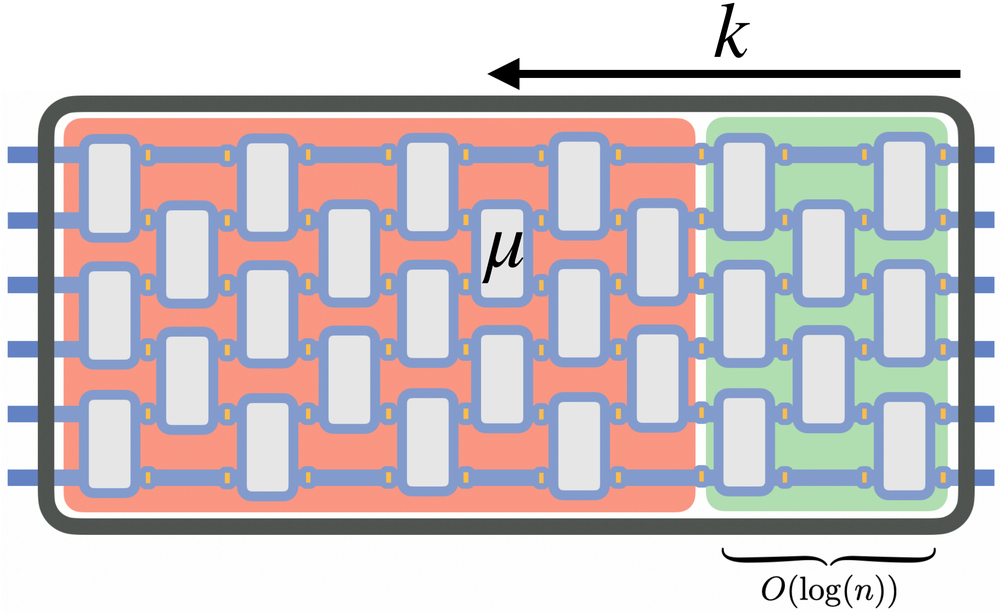
- Cost functions are made by Pauli expectation values: $C := \mathrm{Tr}(P\Phi(\rho_0))$

- Do variations in the gate μ influence C ? Relevant quantity is $\partial_\mu C$.

If $\mathrm{Var}[\partial_\mu C] \leq \exp(-\Theta(n))$, then the influence is very unlikely.

(a.k.a., the gate is **not trainable**)

$\Phi =$



Cost function:

$$C := \mathrm{Tr}(P\Phi(\rho_0))$$

- **Noiseless** [1]**:** Barren plateaus for depth $\geq \Omega(n)$.

(All the gates are not-trainable)

$$\mathrm{Var}[C] \leq \exp(-\Theta(n)),$$
$$\mathrm{Var}[\partial_\mu C] \leq \exp(-\Theta(n))\,.$$

- **Depolarizing noise** [2]**:** Same of before.

- **Non-unital noise [This work]:** No Barren plateaus at any depth for local cost functions.

Only the last $\Theta(\log(n))$ layers are trainable!

$$\mathrm{Var}[C] = \Omega(1),$$
$$\mathrm{Var}[\partial_\mu C] = \exp(-\Theta(k))\,.$$

[1] Barren plateaus in quantum neural network training landscapes. McClean *et al.* Nature Comm. (2018).
[2] Noise-induced barren plateaus in variational quantum algorithms. Wang *et al,* Nature Comm. (2021).
Compare with: [3] Beyond unital noise in variational quantum algorithms: noise-induced barren plateaus and fixed points. Singkanipa *et al.,* ArXiv. (2024).

(Lack of) Cost concentration

Theorem.

Let $H := \sum_{P \in \{I,X,Y,Z\}^{\otimes n} \setminus I^{\otimes n}} a_P P$ be an Hamiltonian, ρ_0 an initial state, for any depth of the non-unital noisy circuit Φ, we have:

$$\mathbb{E}_{\Phi}[\mathrm{Tr}(H\Phi(\rho_0))] = 0,$$

$$\mathrm{Var}_{\Phi}[\mathrm{Tr}(H\Phi(\rho_0))] = \sum_{P \in \{I,X,Y,Z\}^{\otimes n} \setminus I^{\otimes n}} a_P^2 \exp(-\Theta(|P|)),$$

where $|P|$ is the Pauli weight.

Corollaries: - Local cost functions do not suffer from exponential concentration.

(For example, $\mathrm{Var}_{\Phi}[\mathrm{Tr}(Z_1\Phi(\rho_0))] = \Omega(1)$.)

- Global cost functions suffer from exponential concentration.

(For example, $\mathrm{Var}_{\Phi}[\mathrm{Tr}(Z^{\otimes n}\Phi(\rho_0))] = \exp(-\Theta(n))$.)

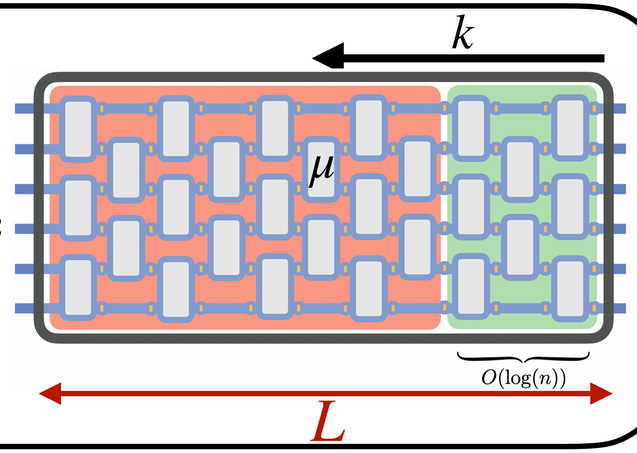
Partial derivatives

Theorem.

Let $C = \mathrm{Tr}(\Phi(\rho_0)P)$, where P is a Pauli, at any depth and noise:

$$\mathbb{E}_{\Phi}[\partial_\mu C] = 0$$

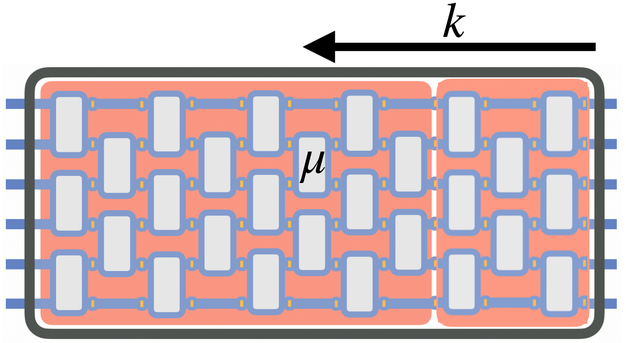
$$\mathrm{Var}_{\Phi}[\partial_\mu C] \leq \exp(-\Theta(k + |P|)),$$

$\Phi =$ 

- If P is local, then for any non-unital noise $\mathrm{Var}_{\Phi}[\partial_\mu C] \geq \exp(-\Theta(k))$.

  Corollary: $\mathbb{E}_{\Phi}[\|\nabla C\|_2^2] = \Omega(1)$.

- If P is global, then for any gate $\mathrm{Var}_{\Phi}[\partial_\mu C] = \exp(-\Theta(n))$.

$\Phi =$ 

- For the particular case of unital noise, our tools imply: $\mathrm{Var}_{\Phi}[\partial_\mu C] \leq \exp(-\Theta(L+|P|))$, which improves upon [1].

[1] Noise-induced barren plateaus in variational quantum algorithms. Wang *et al,* Nature Comm. (2021).

Proof ingredients



- Work in the Heisenberg picture. $\mathrm{Tr}(P\Phi(\rho_0)) = \mathrm{Tr}(\Phi^*(P)\rho_0)$

- 2-design properties (of the 2-qubits gates).

- ‘Normal form’ of noisy channels.

Any 1-qubit channel is unitarily equivalent to a channel parametrized by:

$$(D_X, D_Y, D_Z), (t_X, t_Y, t_Z) \in \mathbb{R}^3$$

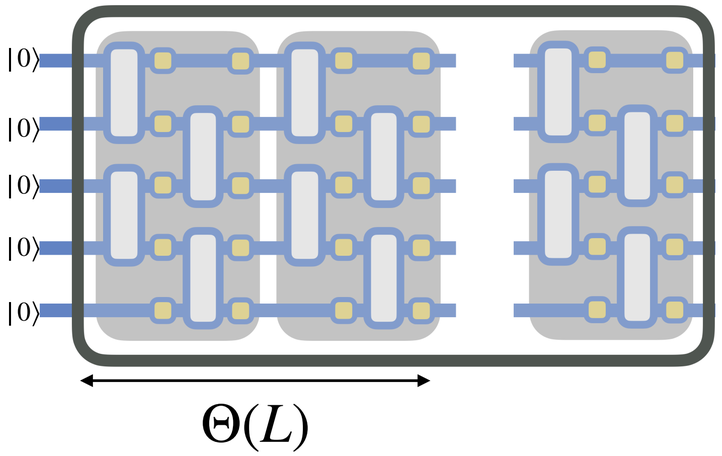
$$\mathcal{N}(I) = I + t_X X + t_Y Y + t_Z Z$$

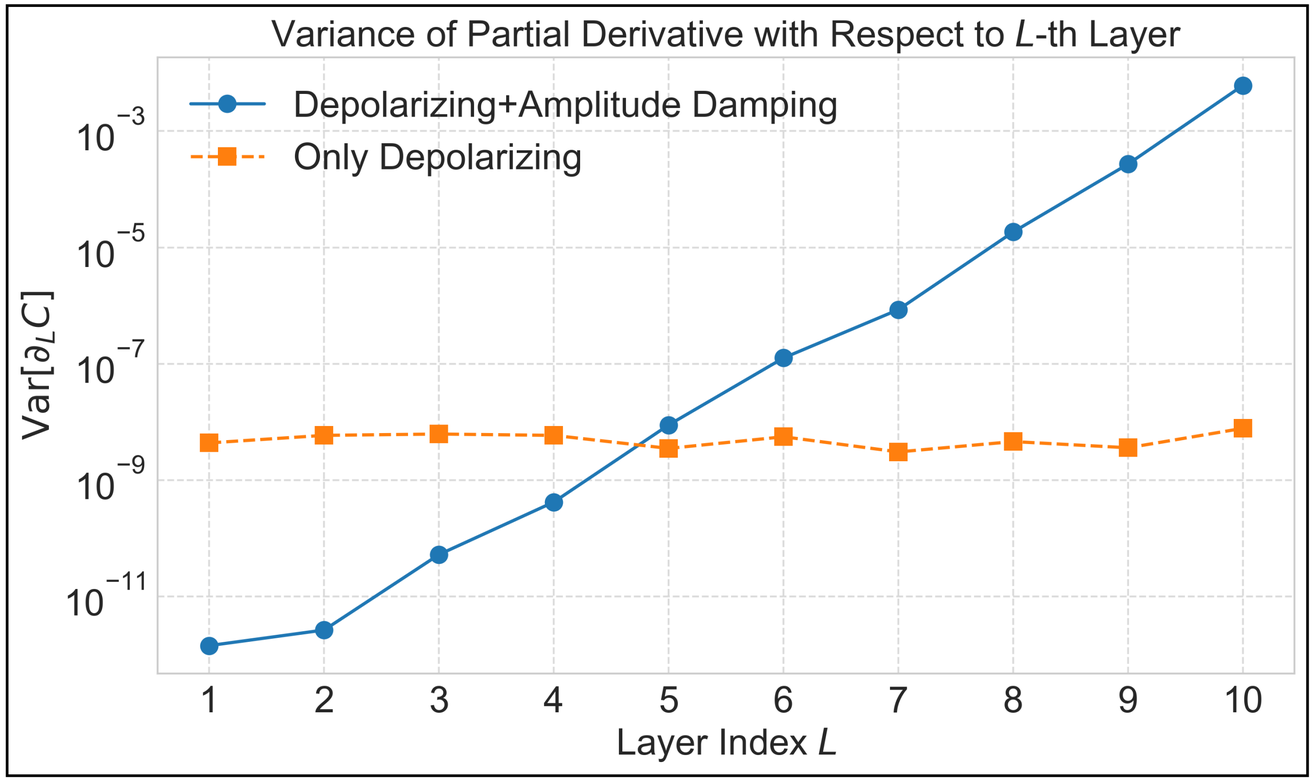
$$\mathcal{N}(X) = D_X X$$

$$\mathcal{N}(Y) = D_Y Y$$

$$\mathcal{N}(Z) = D_Z Z$$

Numerical experiments confirm our results for more structured circuits.





Conclusions

- Any amount of **non-unital noise induces lack of Barren plateaus**, but it 'truncates' most circuits to **effectively log-depth**, allowing **efficient classical simulation** in estimating Pauli expectation values.
- **Unless** we carefully **engineer the circuits to take advantage of the noise**, it is **unlikely** that **non-unital** noisy circuits are **preferable over depolarizing** noisy circuits (which are restricted to log depth).
- Assuming overly simplistic noise models could be misleading.

Open questions

- Simulating Pauli expectation values of noisy random quantum circuits with $\mathrm{poly}(n^{-1})$ accuracy in 2D and all-to-all connectivity in polynomial time.
- Complexity of classical simulation of random quantum circuit sampling with non-unital noise [1].

(The depolarizing case was addressed in [2])

Thanks a lot for your attention!

[1] Effect of non-unital noise on random circuit sampling, Fefferman *et al*, QIP (2023).
[2] A polynomial-time classical algorithm for noisy random circuit sampling, Aharonov *et al*., STOC (2023)